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Using the DeGerlia Time Dilation (DTD) Framework to make Relative Gravitational Time Dilation (GTD) Calculations

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ABSTRACT:In this paper, we reveal that the time dilation between two systems is equal to the ratio of their respective DeGerlia compactness D is defined as m/mean r for about the axis of observation. Fundamental to this exploration is the recognition that relativity is always relativistic, meaning there's no single privileged frame of observation. No measurement, whether time or anything else, has any meaning without a comparator. We reveal this by taking our relative compactness equation plugging in a system compared to a that same system at schwarschild raius and demonstrating that we get the same exact result as you would get doing the gravitational time dilation equation. In order to present this equivalence and demonstrate the relativistic principle, we utilize a framework developed by the author called the DeGerlia time dilation framework, which is a simple, mathematically equivalent time dilation framework that is directly derived from general relativity.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

Under general relativity, The relative pace of time or "time dilation" between two systems is the ratio of their respective mass over radius ratio. We demonstrate this herein by comparing a variety of systems across a range of scenarios from equivalent mass, equivalent radius, equivalent density, equivalent inertia and equivalent density. Mass, radius, density and inertia. And then we also do a general case where we took systems that were distinct across both of those properties. We did systems ranging from classical systems through cosmic systems. So ones where there was almost no relativistic effect and ones where there was extreme relativistic effect. And from this demonstrate mathematical equivalence between time dilation calculated in this manner and in the traditional gravitational time dilation formula.

Key to this exploration is the notion of relativity; that no measurement has any meaning in the absence of a comparator. The Theory of Relativity not only states this in name but it also explains that the way something is observed is influenced by the frame from which it was observed. That there is no privileged frame from which to observe. For example, if you're running fast, you observe yourself passing others who are running slower than you. If you were running slowly, you would observe others passing you that were running faster than you. The same person is observed as fast from one frame and as slow from the other.

To exemplify this notion further, The gravitational time dilation formula takes only a single mass and radius as input, and returns a value that is the gravitational time dilation relative to flat space-time. Most people don't see this as a comparison, however, closer inspection reveals the calculation is comparing the input system with a system of equal mass at its Schwarzschild radius. Gravitational time dilation is described is "the pace of time relative to flat space-time", which is also misleading, considering that flat space-time is an asymptotic limit and not something that's ever reached.

To complete this exploration, we utilize a framework, developed by the author called the DeGerlia Time Dilation Framework. This mathematical complement to general relativity is derived directly from Einstein's equations, while offering several advantages: greatly simplified and more intuitive calculations, is absent of imaginary numbers, and it treats Schwarzschild radius as a point of reference at one equal to number one; versus conflating it to be the singularity. These challenges described are not failings of relativity but rather failings of its interpretation.

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is perceived This paper demonstrates that general and special relativity are always comparative models, and that the very structure of general relativity reveals much about this princi-

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ple that is obscured behind the conventional, sorry, behind the Schwarzschild solution, for lack of a better word. We demonstrate herein a direct relationship between time dilation between two systems is the natural model for evaluating things under general relativity. And in fact the Schwarzschild metric that we currently use for gravitational collapse is just a two system comparison in disguise. To describe it succinctly, The Schwarzschild solution is simply two systems. One being the system you're evaluating, and the other being a system of identical mass at its own Schwarzschild radius. This calculation is understood to produce the relative pace of time between the system being evaluated and what is referred to as "flat space-time" (essentially zero time dilation relative to the system). We show that time, or rather, the pace of time for a system, is emergent of that system's relaive spatial distribution of mass. That spatial distribution is measured axially in units of mass over mean radius, refered to herein as DeGerlia Compactness. Further, we demonstrate that not only is this formula predictive of static mass comparisons such as the Schwarzschild condition. It is also universally predictive of other constrained systems we explore herein as well as a universal comparison of two systems of any scale or compactness. Finally begin to introduce a few important related principles that we will expand upon in future papers.

General relativity is a model for observing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a a product of the systems respective geometric mass distribution. Directly from the gravittional time dilation formula one can derive the underlying two-system logic and ultimately GTD is simply a comparison between a system a systems pace of time relative to the pace of time of an arbitrarily selected benchmark system, which in this case is the same system at its Schwarzschild radius. The formula allows you the freedom to only insert one mass and one radius, and the second is derived directly from the first. This has the effect of hiding the true nature of relativity and that is that it is, above all, comparative. But sends a signal that one need not compare two systems to understand time, and that is flawed thinking; it introduces the mathematical fallacy of an arbitrarily selected comparator, when there is, in fact, no privilege perspective from which to understand time. In this paper we evaluate the physical property responsible for relativistic time dilation and derive the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is dictated by their relative compactness. The standard formula for compactness, described under general relativity, expresses this as a dimensionless parameter. However, in this paper we utilize a principle developed by the author, DeGerlia Compactness, which is an abstracted compactness measure explained herein, that retains the proper unit of this property and that is mass over radius. Thus DeGerlia compactness is defined as a system's mass over mean radius $D = m/\bar{r}$[1]. We do a comparative analysis and tabulate relative time dilation across a variety of systems from classical to relativistic regimes, and show that the same compactness ratio $k = D_1/D_2 = M_1R_2/(M_2R_1)$ returns the Schwarzschild time-dilation ratio exactly across every tested constraint class, including the fully general case. This establishes that the relative rate of time between systems is not an ad hoc relativistic output requiring separate formulas, but a direct consequence of comparative mass-radius geometry. The result makes explicit a simple relation already contained in general relativity: given two systems' masses and mean radii, their relative time dilation is determined by their respective geometric distribution of mass alone.

we evaluate the nature of general relativity and expose a few important considerations for gravitational time dilation calculations. First and foremost, relativity describes the relative nature of the universe; thst no quantity has physical meaning without a pgysical comparator. This is true of time, and the pace of time for a system is only meaningful relative to the pace of time of another system. The standard formula for gravitational time dilation (GTD) is misleading in that it sends the signal that GTD can be calculated for a single system without a comparator, and that the comparator is flat space-time, which is not actually a thing. In this paper we demonstrate that GTD is always and only a comparison between two systems, and that the relative pace of time between those systems emerges exclusively from the spatial distribution of mass in the respective systems, from the respective inertial density of those systems. To demonstrate this, we use a simple, mathematically equivalent time-dilation framework developed by the author, referred to as the DeGerlia Time Dilation Framework, to explore several constrained system examples. We conclude by validating the calculations against gravitational time dilation calculations based on Schwarzschild radius ratios.

In this paper, we demonstrate that the relative pace of time between two systems, or time dilation between those systems, emerges exclusively from the spatial distribution of mass in the respective systems, from the respective inertial density of those systems. To demonstrate this, we use a simple, mathematically equivalent time-dilation framework developed by the author, referred to as the DeGerlia Time Dilation Framework, to explore several constrained system examples. We conclude by validating the calculations against gravitational time dilation

System time emerges from spatial distribution of mass relative to that of another system. Though General Relativity spells it out clearly, this phenomenon is commonly described as the curvature of space-time as influenced by the presence of a nearby mass. That statement, in and of itself, is flawed in several respects. a) While the clock time ratio can be considered a continuum, it's an asymptotic approach, not a limit that's ever reached. b) While the clock time ratio can range from 0 to 1, both limits are asymptotic approaches to those numbers.

Relativity is always and only comparative. The single-system formula used most often for gravitational time dilation is misleading and inaccurate. It sends the signal that gravity has meaning without a comparator. As such, it builds in an arbitrarily selected comparator: Flat Space-Time, which is not actually a thing. This misconception would have significant effects on values calculated closer to the Schwarzschild radius.

We use the DeGerlia framework to calculate gravitational time dilation based on relative inertial density properties. This demonstrates that because we're able to take my 2-system equation and run it with one system being the Schwarzschild radius and the radius of the first system. This validates both the comparative nature of relativity and the applicability of the DeGerlia framework.

Several distinct, common forms of time dilation utilize the same formula. Utilize a common formula, and that is that the time dilation is equal to $\sqrt{1-k}$, the square root of one minus the scale factor. In general relativity, the scale factor is calculated by r_s/R. In special relativity, it's calculated as v^2/c^2, and in general relativity, you can also substitute that: r_s/R could be replaced with D/D_{crit}. Those are all equivalent.

Both General and Special Relativity follow a very similar form, and their distinction mirrors the distinction between linear and rotational inertia. This alludes to a very rational and logical unification of relativity.

2 Symbol Glossary

Symbol	Definition / value
GTD	$\sqrt{1-k} = d\tau/dt$
DTD	$\sqrt{k}$ (DeGerlia time dilation)
k	$v^2/c^2 = r_s/R = D/D_{\text{crit}}$
D	$M/\bar{R}_{\text{axis}}$ (DeGerlia compactness)
D_{crit}	$c^2/2G = 6.73295 \times 10^{26}$ kg/m
r_s	$2GM/c^2$ (Schwarzschild radius)
P	I/V (inertial density) $= D \cdot 3/10\pi$
θ	$\arcsin\sqrt{k}$; GTD $= \cos\theta$, DTD $= \sin\theta$
c	299,792,458 m/s (exact)
G	6.67430×10^{-11} m^{3}kg^{-1}s^{-2}

3 General Relativity and Compactness

3.1 Compactness

Compactness is the dimensionless ratio of a system's mass to its radius, scaled by G/c^2. The most common forms are $C = GM/(Rc^2)$ and $r_s/R = 2GM/(Rc^2) = 2C$, where $r_s = 2GM/c^2$ is the Schwarzschild radius. Both measure how close a system sits to its Schwarzschild threshold: $r_s/R = 1$ marks the gravitational-collapse condition.

3.2 Time Dilation

3.3 Structure of the GTD Equation as a Two-System Comparison

4 The DeGerlia Time Dilation Framework

An analogous framework that is directly derived from general relativity and has a direct relationship to gravitational time dilation.

4.1 The Scale Factor k

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{1}$$

* This identity has been previously referred to as the "space-time equivalence"[2].

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i. The three forms in Equation 1 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1R_2/(M_2R_1)$ is the fully expanded form.

A system's pace of time has meaning only relative to another pace of time. The scale factor k governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k}$, where $k = D/D_{crit} = r_s/R$. The compactness ratio $k = D_1/D_2$ and the Schwarzschild form give the same value.

Via the Schwarzschild radius, the same scale factor is:

$$k = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{2}$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. This is the k that appears in the standard $\sqrt{1-k}$ for GTD.

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{3}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia compactness ratio for the case being computed. The specific forms are:

- $k = D_1/D_2$ (Equation 1) — via inertial density ratio
- $k = D/D_{crit} = r_s/R$ (Equation 2) — via Schwarzschild radius

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4.2 DeGerlia Compactness D

Inertial density P[1] is the moment of inertia of a system divided by its volume:

$$P = I/V \tag{4}$$

Inertia, in a linear context, is simply mass. Inertial density, in this context, reduces directly to mass density $\rho = M/V$. From this perspective, inertial density is simply the rotational analog to mass density, not unlike the relationship between f equals m a and torque equals angular acceleration times m. More succinctly stated: mass density is simply the linear interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \tag{5}$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

4.3 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)[3] and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26}\ \mathrm{kg/m} \tag{6}$$

Note this value is exact; its precision is limited by $G = (6.67430 \pm 0.00015) \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$[4].

4.4 DeGerlia Time Dilation

DeGerlia Time Dilation is the geometric complement of gravitational time dilation:

$$\mathrm{DTD} = \sqrt{k} \tag{7}$$

where k is the scale factor of Equation 1. Where $\mathrm{GTD} = \sqrt{1-k}$ is the surviving proper-time rate $d\tau/dt$, $\mathrm{DTD} = \sqrt{k}$ is its complement.

4.5 Converting Between GTD and DTD

GTD and DTD are bound by the Pythagorean identity:

$$\mathrm{GTD}^2 + \mathrm{DTD}^2 = (1-k) + k = 1. \tag{8}$$

4.6 Properties of DTD

5 Relative Time Dilation Examples

Example parameter sets match the columns of the pair comparison tables in Appendix A.

5.1 Static Mass ($M_1 = M_2$)

When the two systems share the same mass, the mass terms cancel in k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This example compares a system at $\frac{4}{3} r_s$, where $\mathrm{GTD} = \frac{1}{2}$, against a system at an ordinary radius far from its Schwarzschild radius.

Example (Static Mass):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e30
R (m)	1.98031e3	1.00000e8

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(R D_{\mathrm{crit}})$$
$$K_1 = \frac{1.00000\mathrm{e}30\,\mathrm{kg}}{(1.98031\mathrm{e}3\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 7.50001\mathrm{e}{-1}$$
$$\mathrm{DTD}_1 = \sqrt{7.50001\mathrm{e}{-1}} = 8.66026\mathrm{e}{-1}$$
$$K_2 = \frac{1.00000\mathrm{e}30\,\mathrm{kg}}{(1.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-5}$$
$$\mathrm{DTD}_2 = \sqrt{1.48523\mathrm{e}{-5}} = 3.85387\mathrm{e}{-3}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{8.66026\mathrm{e}{-1}}{3.85387\mathrm{e}{-3}} = 2.24716\mathrm{e}2$$

Calculate GTD ratio from DTD:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}} = \frac{\sqrt{1-(8.66026\mathrm{e}{-1})^2}}{\sqrt{1-(3.85387\mathrm{e}{-3})^2}} = \frac{\sqrt{1-7.50001\mathrm{e}{-1}}}{\sqrt{1-1.48523\mathrm{e}{-5}}} = \frac{\sqrt{2.49999\mathrm{e}{-1}}}{\sqrt{9.999851477\mathrm{e}{-1}}} = \frac{4.99999\mathrm{e}{-1}}{9.9999257381\mathrm{e}{-1}} = 5.00003\mathrm{e}{-1} = 1 - 4.99997\mathrm{e}{-1}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2} = 1.48523\mathrm{e}3\,\mathrm{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2} = 1.48523\mathrm{e}3\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\mathrm{e}3\,\mathrm{m}}{1.98031\mathrm{e}3\,\mathrm{m}} = 7.50000\mathrm{e}{-1}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\mathrm{e}3\,\mathrm{m}}{1.00000\mathrm{e}8\,\mathrm{m}} = 1.48523\mathrm{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1 = \sqrt{1-K_1} = \sqrt{2.50000\mathrm{e}{-1}} = 5.00000\mathrm{e}{-1} = 1 - 5.00000\mathrm{e}{-1}$$
$$\mathrm{GTD}_2 = \sqrt{1-K_2} = \sqrt{9.999851477\mathrm{e}{-1}} = 9.9999257381\mathrm{e}{-1} = 1 - 7.42619\mathrm{e}{-6}$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{5.00000\mathrm{e}{-1}}{9.9999257381\mathrm{e}{-1}} = 5.00004\mathrm{e}{-1} = 1 - 4.99996\mathrm{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 5.00004\mathrm{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 5.00003\mathrm{e}{-1} \quad (\text{via } D_{\mathrm{crit}})$$

5.2 Example (Static Mass, Table Case 1)

In the following example the reference system is placed at $1.00001\,r_s$ rather than exactly at its Schwarzschild radius. Both limits of the gravitational time-dilation range are asymptotic: $R \to r_s$ gives $\mathrm{GTD} \to 0$ and $R \to \infty$ gives $\mathrm{GTD} \to 1$. Neither end-point is an admissible input — just as the flat-spacetime reference is approximated by an arbitrarily large finite radius rather than $R=\infty$, the Schwarzschild condition is approximated here by a radius an arbitrarily small increment above r_s, yielding a finite value while the offset remains below the sixth significant figure.

Example (Static Mass):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000e30	2.00000e30
R (m)	2.97049e3	7.00000e8

Calculate DTD for each System:

$$\mathrm{DTD}=\sqrt{K}\quad\text{where}\quad K=M/(RD_{\mathrm{crit}})$$
$$K_1=\frac{2.00000\mathrm{e}30\,\mathrm{kg}}{(2.97049\mathrm{e}3\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})}=9.9999068441\mathrm{e}{-1}$$
$$\mathrm{DTD}_1=\sqrt{9.9999068441\mathrm{e}{-1}}=9.9999534220\mathrm{e}{-1}$$
$$K_2=\frac{2.00000\mathrm{e}30\,\mathrm{kg}}{(7.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})}=4.24352\mathrm{e}{-6}$$
$$\mathrm{DTD}_2=\sqrt{4.24352\mathrm{e}{-6}}=2.05998\mathrm{e}{-3}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2}=\frac{9.9999534220\mathrm{e}{-1}}{2.05998\mathrm{e}{-3}}=4.85439\mathrm{e}2$$

Calculate GTD ratio from DTD:

$$\begin{aligned}\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2}&=\frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}}\\&=\frac{\sqrt{1-(9.9999534220\mathrm{e}{-1})^2}}{\sqrt{1-(2.05998\mathrm{e}{-3})^2}}\\&=\frac{\sqrt{1-9.9999068441\mathrm{e}{-1}}}{\sqrt{1-4.24352\mathrm{e}{-6}}}\\&=\frac{\sqrt{9.315586\mathrm{e}{-6}}}{\sqrt{9.9999575648\mathrm{e}{-1}}}\\&=\frac{3.05214\mathrm{e}{-3}}{9.9999787824\mathrm{e}{-1}}\\&=3.05215\mathrm{e}{-3}=1-9.9694785\mathrm{e}{-1}\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1}=\frac{2GM_1}{c^2}=\frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(2.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2}$$
$$=2.97046\mathrm{e}3\,\mathrm{m}$$
$$r_{s2}=\frac{2GM_2}{c^2}=\frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(2.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2}$$
$$=2.97046\mathrm{e}3\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1=\frac{r_{s1}}{R_1}=\frac{2.97046\mathrm{e}3\,\mathrm{m}}{2.97049\mathrm{e}3\,\mathrm{m}}=9.9999000010\mathrm{e}{-1}$$
$$K_2=\frac{r_{s2}}{R_2}=\frac{2.97046\mathrm{e}3\,\mathrm{m}}{7.00000\mathrm{e}8\,\mathrm{m}}=4.24352\mathrm{e}{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1=\sqrt{1-K_1}=\sqrt{9.9999000010\mathrm{e}{-6}}$$
$$=3.16226\mathrm{e}{-3}=1-9.9683774\mathrm{e}{-1}$$
$$\mathrm{GTD}_2=\sqrt{1-K_2}=\sqrt{9.9999575648\mathrm{e}{-1}}$$
$$=9.9999787824\mathrm{e}{-1}=1-2.12176\mathrm{e}{-6}$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2}=\frac{3.16226\mathrm{e}{-3}}{9.9999787824\mathrm{e}{-1}}$$
$$=3.16227\mathrm{e}{-3}=1-9.9683773\mathrm{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2}=3.16227\mathrm{e}{-3}\quad(\text{via } r_s)$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2}=3.05215\mathrm{e}{-3}\quad(\text{via } D_{\mathrm{crit}})$$

5.3 Example - Static Mass , Table 2 reference M=M Case

When the two systems share the same mass, the mass terms cancel in k_d:

$$k_d=\frac{M_1R_2}{M_2R_1}=\frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D=M/R$, this is $D_1/D_2=R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit}=r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This case is the standard gravitational time-dilation calculation; the $M_1R_2/(M_2R_1)$ ratio reduces to the familiar Schwarzschild factor here. Both systems are written out and computed independently below — the same Schwarzschild calculation, with the second system (the same mass at its Schwarzschild radius) made visible instead of folded into r_s.

Example (Static Mass):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000e30	2.00000e30
R (m)	3.00000e3	7.00000e8

Calculate DTD for each System:

$$\mathrm{DTD}=\sqrt{K}\quad\text{where}\quad K=M/(RD_{\mathrm{crit}})$$
$$K_1=\frac{2.00000\mathrm{e}30\,\mathrm{kg}}{(3.00000\mathrm{e}3\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})}=9.9015538\mathrm{e}{-1}$$
$$\mathrm{DTD}_1=\sqrt{9.9015538\mathrm{e}{-1}}=9.9506552\mathrm{e}{-1}$$
$$K_2=\frac{2.00000\mathrm{e}30\,\mathrm{kg}}{(7.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})}=4.24352\mathrm{e}{-6}$$
$$\mathrm{DTD}_2=\sqrt{4.24352\mathrm{e}{-6}}=2.05998\mathrm{e}{-3}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2}=\frac{9.9506552\mathrm{e}{-1}}{2.05998\mathrm{e}{-3}}=4.83046\mathrm{e}2$$

Calculate GTD ratio from DTD:

$$\begin{aligned}\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2}&=\frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}}\\&=\frac{\sqrt{1-(9.9506552\mathrm{e}{-1})^2}}{\sqrt{1-(2.05998\mathrm{e}{-3})^2}}\\&=\frac{\sqrt{1-9.9015538\mathrm{e}{-1}}}{\sqrt{1-4.24352\mathrm{e}{-6}}}\\&=\frac{\sqrt{9.844620\mathrm{e}{-3}}}{\sqrt{9.9999575648\mathrm{e}{-1}}}\\&=\frac{9.9220058\mathrm{e}{-2}}{9.9999787824\mathrm{e}{-1}}\\&=9.9220268\mathrm{e}{-2}=1-9.007797\mathrm{e}{-1}\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1}=\frac{2GM_1}{c^2}=\frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(2.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2}$$
$$=2.97046\mathrm{e}3\,\mathrm{m}$$
$$r_{s2}=\frac{2GM_2}{c^2}=\frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(2.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2}$$
$$=2.97046\mathrm{e}3\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1=\frac{r_{s1}}{R_1}=\frac{2.97046\mathrm{e}3\,\mathrm{m}}{3.00000\mathrm{e}3\,\mathrm{m}}=9.9015470\mathrm{e}{-1}$$
$$K_2=\frac{r_{s2}}{R_2}=\frac{2.97046\mathrm{e}3\,\mathrm{m}}{7.00000\mathrm{e}8\,\mathrm{m}}=4.24352\mathrm{e}{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{9.845297\mathrm{e}{-3}} = 9.9223472\mathrm{e}{-2} \\
&= 1 - 9.007765\mathrm{e}{-1} \\
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{9.9999575648\mathrm{e}{-1}} \\
&= 9.9999787824\mathrm{e}{-1} = 1 - 2.12176\mathrm{e}{-6} \\
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{9.9223472\mathrm{e}{-2}}{9.9999787824\mathrm{e}{-1}} \\
&= 9.9223683\mathrm{e}{-2} = 1 - 9.007763\mathrm{e}{-1}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 9.9223683\mathrm{e}{-2} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 9.9220268\mathrm{e}{-2} \quad (\text{via } D_{\mathrm{crit}})$$

5.4 Static Radius (Linear) ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{M_1}{M_2}$$

The DeGerlia compactness ratio reduces to a ratio of masses. The Lorentz factor has the same form as Equation 3, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e24	1.00000e30
R (m)	1.00000e8	1.00000e8

Calculate DTD for each System:

$$\begin{aligned}
\mathrm{DTD} &= \sqrt{K} \quad \text{where} \quad K = M/(R D_{\mathrm{crit}}) \\
K_1 &= \frac{1.00000\mathrm{e}24\,\mathrm{kg}}{(1.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-11} \\
\mathrm{DTD}_1 &= \sqrt{1.48523\mathrm{e}{-11}} = 3.85387\mathrm{e}{-6} \\
K_2 &= \frac{1.00000\mathrm{e}30\,\mathrm{kg}}{(1.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-5} \\
\mathrm{DTD}_2 &= \sqrt{1.48523\mathrm{e}{-5}} = 3.85387\mathrm{e}{-3} \\
\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} &= \frac{3.85387\mathrm{e}{-6}}{3.85387\mathrm{e}{-3}} = 1.00000\mathrm{e}{-3}
\end{aligned}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_2}{\mathrm{GTD}_1} &= \frac{\sqrt{1-\mathrm{DTD}_2{}^2}}{\sqrt{1-\mathrm{DTD}_1{}^2}} \\
&= \frac{\sqrt{1-(3.85387\mathrm{e}{-3})^2}}{\sqrt{1-(3.85387\mathrm{e}{-6})^2}} \\
&= \frac{\sqrt{1-1.48523\mathrm{e}{-5}}}{\sqrt{1-1.48523\mathrm{e}{-11}}} \\
&= \frac{\sqrt{9.999851477\mathrm{e}{-1}}}{\sqrt{9.999999999851477\mathrm{e}{-1}}} \\
&= \frac{9.9999257381\mathrm{e}{-1}}{9.9999999999925738\mathrm{e}{-1}} \\
&= 9.9999257381\mathrm{e}{-1} = 1 - 7.42619\mathrm{e}{-6}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000\mathrm{e}24\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523\mathrm{e}{-3}\,\mathrm{m} \\
r_{s2} &= \frac{2GM_2}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523\mathrm{e}3\,\mathrm{m}
\end{aligned}$$

Calculate K values from the r_s/R ratio:

$$\begin{aligned}
K_1 &= \frac{r_{s1}}{R_1} = \frac{1.48523\mathrm{e}{-3}\,\mathrm{m}}{1.00000\mathrm{e}8\,\mathrm{m}} = 1.48523\mathrm{e}{-11} \\
K_2 &= \frac{r_{s2}}{R_2} = \frac{1.48523\mathrm{e}3\,\mathrm{m}}{1.00000\mathrm{e}8\,\mathrm{m}} = 1.48523\mathrm{e}{-5}
\end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{9.999999999851477\mathrm{e}{-1}} \\
&= 9.9999999999925738\mathrm{e}{-1} \\
&= 1 - 7.42616\mathrm{e}{-12} \\
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{9.999851477\mathrm{e}{-1}} \\
&= 9.9999257381\mathrm{e}{-1} = 1 - 7.42619\mathrm{e}{-6} \\
\frac{\mathrm{GTD}_2}{\mathrm{GTD}_1} &= \frac{9.9999257381\mathrm{e}{-1}}{9.9999999999925738\mathrm{e}{-1}} \\
&= 9.9999257382\mathrm{e}{-1} = 1 - 7.42618\mathrm{e}{-6}
\end{aligned}$$

Time-dilation ratio between System 2 and System 1, computed two ways:

$$\frac{\mathrm{GTD}_2}{\mathrm{GTD}_1} = 9.9999257382\mathrm{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_2}{\mathrm{GTD}_1} = 9.9999257381\mathrm{e}{-1} \quad (\text{via } D_{\mathrm{crit}})$$

5.5 Constant Density ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The DeGerlia compactness ratio reduces to the square of the radius ratio; the linear scale factor follows as $\sqrt{k_d} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e27
R (m)	1.00000e8	1.00000e7

Calculate DTD for each System:

$$\begin{aligned}
\mathrm{DTD} &= \sqrt{K} \quad \text{where} \quad K = M/(R D_{\mathrm{crit}}) \\
K_1 &= \frac{1.00000\mathrm{e}30\,\mathrm{kg}}{(1.00000\mathrm{e}8\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-5} \\
\mathrm{DTD}_1 &= \sqrt{1.48523\mathrm{e}{-5}} = 3.85387\mathrm{e}{-3} \\
K_2 &= \frac{1.00000\mathrm{e}27\,\mathrm{kg}}{(1.00000\mathrm{e}7\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-7} \\
\mathrm{DTD}_2 &= \sqrt{1.48523\mathrm{e}{-7}} = 3.85387\mathrm{e}{-4} \\
\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} &= \frac{3.85387\mathrm{e}{-3}}{3.85387\mathrm{e}{-4}} = 1.00000\mathrm{e}1
\end{aligned}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}} \\
&= \frac{\sqrt{1-(3.85387\mathrm{e}{-3})^2}}{\sqrt{1-(3.85387\mathrm{e}{-4})^2}}
\end{aligned}$$

$$= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-7}}}$$
$$= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.99999851477\text{e}{-1}}}$$
$$= \frac{9.9999257381\text{e}{-1}}{9.999999257383\text{e}{-1}}$$
$$= 9.99992648 07\text{e}{-1} = 1 - 7.35193\text{e}{-6}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}27\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\,\text{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-7}$$

Calculate GTD via Schwarzschild radius ratio:

$$\text{GTD}_1 = \sqrt{1-K_1} = \sqrt{9.999851477\text{e}{-1}}$$
$$= 9.9999257381\text{e}{-1} = 1 - 7.42619\text{e}{-6}$$
$$\text{GTD}_2 = \sqrt{1-K_2} = \sqrt{9.99999851477\text{e}{-1}}$$
$$= 9.999999257384\text{e}{-1} = 1 - 7.42616\text{e}{-8}$$
$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{9.9999257381\text{e}{-1}}{9.999999257384\text{e}{-1}}$$
$$= 9.99992648 07\text{e}{-1} = 1 - 7.35193\text{e}{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = 9.9999264807\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\text{GTD}_1}{\text{GTD}_2} = 9.9999264807\text{e}{-1} \quad (\text{via } D_{\text{crit}})$$

5.6 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1R_1^2 = M_2R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_d:

$$k_d = \frac{M_1R_2}{M_2R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e32	1.00000e30
R (m)	1.00000e7	1.00000e8

Calculate DTD for each System:

$$\text{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(RD_{\text{crit}})$$
$$K_1 = \frac{1.00000\text{e}32\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-2}$$
$$\text{DTD}_1 = \sqrt{1.48523\text{e}{-2}} = 1.21870\text{e}{-1}$$
$$K_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$\text{DTD}_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$\frac{\text{DTD}_1}{\text{DTD}_2} = \frac{1.21870\text{e}{-1}}{3.85387\text{e}{-3}} = 3.16228\text{e}1$$

Calculate GTD ratio from DTD:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{\sqrt{1-\text{DTD}_1{}^2}}{\sqrt{1-\text{DTD}_2{}^2}}$$
$$= \frac{\sqrt{1-(1.21870\text{e}{-1})^2}}{\sqrt{1-(3.85387\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-1.48523\text{e}{-2}}}{\sqrt{1-1.48523\text{e}{-5}}}$$
$$= \frac{\sqrt{9.851477\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}}$$
$$= \frac{9.9254605\text{e}{-1}}{9.9999257381\text{e}{-1}}$$
$$= 9.9255342\text{e}{-1} = 1 - 7.44658\text{e}{-3}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}32\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}5\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}5\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-2}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$\text{GTD}_1 = \sqrt{1-K_1} = \sqrt{9.851477\text{e}{-1}} = 9.9254606\text{e}{-1}$$
$$= 1 - 7.45394\text{e}{-3}$$
$$\text{GTD}_2 = \sqrt{1-K_2} = \sqrt{9.999851477\text{e}{-1}}$$
$$= 9.9999257381\text{e}{-1} = 1 - 7.42619\text{e}{-6}$$
$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{9.9254606\text{e}{-1}}{9.9999257381\text{e}{-1}}$$
$$= 9.9255343\text{e}{-1} = 1 - 7.44657\text{e}{-3}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = 9.9255343\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\text{GTD}_1}{\text{GTD}_2} = 9.9255342\text{e}{-1} \quad (\text{via } D_{\text{crit}})$$

5.7 General Case (unconstrained)

When no constraint is applied between the systems, k_d stays in its full form:

$$k_d = \frac{M_1R_2}{M_2R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e20	1.00000e30
R (m)	2.00000e−7	1.00000e7

Calculate DTD for each System:

$$\text{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(RD_{\text{crit}})$$
$$K_1 = \frac{1.00000\text{e}20\,\text{kg}}{(2.00000\text{e}{-7}\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.42617\text{e}{-1}$$
$$\text{DTD}_1 = \sqrt{7.42617\text{e}{-1}} = 8.61752\text{e}{-1}$$
$$K_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-4}$$

$$\mathrm{DTD}_2 = \sqrt{1.48523\mathrm{e}{-4}} = 1.21870\mathrm{e}{-2}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{8.61752\mathrm{e}{-1}}{1.21870\mathrm{e}{-2}} = 7.07107\mathrm{e}1$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}} \\
&= \frac{\sqrt{1-(8.61752\mathrm{e}{-1})^2}}{\sqrt{1-(1.21870\mathrm{e}{-2})^2}} \\
&= \frac{\sqrt{1-7.42617\mathrm{e}{-1}}}{\sqrt{1-1.48523\mathrm{e}{-4}}} \\
&= \frac{\sqrt{2.57383\mathrm{e}{-1}}}{\sqrt{9.99851477\mathrm{e}{-1}}} \\
&= \frac{5.07330\mathrm{e}{-1}}{9.999257356\mathrm{e}{-1}} \\
&= 5.07367\mathrm{e}{-1} = 1-4.92633\mathrm{e}{-1}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(1.00000\mathrm{e}20\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s}^{-1})^2}$$
$$= 1.48523\mathrm{e}{-7}\,\mathrm{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(1.00000\mathrm{e}30\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s}^{-1})^2}$$
$$= 1.48523\mathrm{e}3\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\mathrm{e}{-7}\,\mathrm{m}}{2.00000\mathrm{e}{-7}\,\mathrm{m}} = 7.42616\mathrm{e}{-1}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\mathrm{e}3\,\mathrm{m}}{1.00000\mathrm{e}7\,\mathrm{m}} = 1.48523\mathrm{e}{-4}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{2.57384\mathrm{e}{-1}} = 5.07330\mathrm{e}{-1} \\
&= 1-4.92670\mathrm{e}{-1} \\
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{9.99851477\mathrm{e}{-1}} \\
&= 9.999257356\mathrm{e}{-1} = 1-7.42644\mathrm{e}{-5} \\
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{5.07330\mathrm{e}{-1}}{9.999257356\mathrm{e}{-1}} \\
&= 5.07368\mathrm{e}{-1} = 1-4.92632\mathrm{e}{-1}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 5.07368\mathrm{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 5.07367\mathrm{e}{-1} \quad (\text{via } D_{\mathrm{crit}})$$

5.8 Classical Static Density Example

Example (Classical):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	8.00000	1.00000
R (m)	2.00000	1.00000

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(RD_{\mathrm{crit}})$$
$$K_1 = \frac{8.00000\,\mathrm{kg}}{(2.00000\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 5.94093\mathrm{e}{-27}$$
$$\mathrm{DTD}_1 = \sqrt{5.94093\mathrm{e}{-27}} = 7.70774\mathrm{e}{-14}$$
$$K_2 = \frac{1.00000\,\mathrm{kg}}{(1.00000\,\mathrm{m})(6.73295\mathrm{e}26\,\mathrm{kg/m})} = 1.48523\mathrm{e}{-27}$$
$$\mathrm{DTD}_2 = \sqrt{1.48523\mathrm{e}{-27}} = 3.85387\mathrm{e}{-14}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{7.70774\mathrm{e}{-14}}{3.85387\mathrm{e}{-14}} = 2.00000$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1{}^2}}{\sqrt{1-\mathrm{DTD}_2{}^2}} \\
&= \frac{\sqrt{1-(7.70774\mathrm{e}{-14})^2}}{\sqrt{1-(3.85387\mathrm{e}{-14})^2}} \\
&= \frac{\sqrt{1-5.94093\mathrm{e}{-27}}}{\sqrt{1-1.48523\mathrm{e}{-27}}} \\
&= \frac{\sqrt{9.9999999999999999999999999405907\mathrm{e}{-1}}}{\sqrt{9.9999999999999999999999999851477\mathrm{e}{-1}}} \\
&= \frac{9.9999999999999999999999999702953\mathrm{e}{-1}}{9.99999999999999999999999999257383\mathrm{e}{-1}} \\
&= 9.9999999999999999999999999777215\mathrm{e}{-1} \\
&= 1-2.22785\mathrm{e}{-27}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(8.00000\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s}^{-1})^2}$$
$$= 1.18819\mathrm{e}{-26}\,\mathrm{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\mathrm{e}{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(1.00000\,\mathrm{kg})}{(2.99792\mathrm{e}8\,\mathrm{m\,s}^{-1})^2}$$
$$= 1.48523\mathrm{e}{-27}\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.18819\mathrm{e}{-26}\,\mathrm{m}}{2.00000\,\mathrm{m}} = 5.94093\mathrm{e}{-27}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\mathrm{e}{-27}\,\mathrm{m}}{1.00000\,\mathrm{m}} = 1.48523\mathrm{e}{-27}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} \\
&= \sqrt{9.9999999999999999999999999405907\mathrm{e}{-1}} \\
&= 9.9999999999999999999999999702954\mathrm{e}{-1} \\
&= 1-2.97046\mathrm{e}{-27} \\
\mathrm{GTD}_2 &= \sqrt{1-K_2} \\
&= \sqrt{9.9999999999999999999999999851477\mathrm{e}{-1}} \\
&= 9.99999999999999999999999999257384\mathrm{e}{-1} \\
&= 1-7.42616\mathrm{e}{-28} \\
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{9.9999999999999999999999999702954\mathrm{e}{-1}}{9.99999999999999999999999999257384\mathrm{e}{-1}} \\
&= 9.9999999999999999999999999777215\mathrm{e}{-1} \\
&= 1-2.22785\mathrm{e}{-27}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 9.9999999999999999999999999777215\mathrm{e}{-1} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = 9.9999999999999999999999999777215\mathrm{e}{-1} \quad (\text{via } D_{\mathrm{crit}})$$

6 Analysis

The reformulation is meant to carry three points, and nothing more ambitious. First, it is a pace-of-time framework that is simply easier to work with: expressed in DTD and compactness, comparing two systems is a direct ratio of geometries, with no benchmark, no subtraction-from-one, and — because D_{crit} cancels — no dependence on G. One returns to GTD, the $\sqrt{1-k}$ form, only at the final step, to convert relative pace into relative clock time. Second, general relativity is always two-system; we merely masquerade it as one. Conventional gravitational time dilation looks like a single-system calculation only because it silently fixes the second system — the same mass at its Schwarzschild radius, where DTD $= 1$. Make that benchmark explicit and the two-system structure, present all along, is plain. Third, treating it as one system is a disservice to the broader mathematics: the collapse to a single system rests on presuming that some property of the hidden second system is a fixed, universal given, a presumption that obscures the real structure and perpetuates a misconception that dilutes the core principle. The honest formulation keeps both systems in view and recognizes the single-system shortcut as a convenience, not as the underlying physics.

7 Conclusions

8 Hypotheses of Explanation

9 Further Exploration

Acknowledgments

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity_STE_cases`.

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Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 5: static mass (M=M), static radius (R=R), constant density ($\rho = \rho$), constant inertia (I=I), an unconstrained general case, and an STE example. The three algebraic forms of k in Equation 1 produce identical values in every case, alongside the gravitational time-dilation ratio computed independently from the standard Schwarzschild form.

Table 1 System properties for each pair comparison case.

	$M=M$, $R_1=\frac{4}{3}r_s$	$M=M$	$R=R$	$\rho=\rho$	$I=I$	**General**	**Classical ($\rho=\rho$)**
m_1 (kg)	1.00000e+30	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
m_2 (kg)	1.00000e+30	2.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
r_1 (m)	1.98031e+3	3.00000e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
r_2 (m)	1.00000e+8	7.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	3.07406e+19	1.76839e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	2.38732e+5	1.39203e+3	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	1.56865e+36	7.20000e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	4.00000e+45	3.92000e+47	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	5.04972e+26	6.66667e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	1.00000e+22	2.85714e+21	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
r_{s_1} (m)	1.48523e+3	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s_2} (m)	1.48523e+3	2.97046e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
D_{1_norm}	7.50001e-1	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
D_{2_norm}	1.48523e-5	4.24352e-6	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
k_s	7.50000e-1	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42616e-1	5.94093e-27
k_d	7.50001e-1	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
GTD$_{s,1}$	1-5.00000e-1	1-9.00777e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
GTD$_{s,2}$	1-7.42619e-6	1-2.12176e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28
GTD$_{d,1}$	1-5.00001e-1	1-9.00780e-1	1-7.42619e-6	1-7.42619e-6	1-7.45395e-3	1-4.92670e-1	1-2.97047e-27
GTD$_{d,2}$	1-7.42619e-6	1-2.12176e-6	1-7.42617e-12	1-7.42617e-8	1-7.42619e-6	1-7.42644e-5	1-7.42617e-28

Table 2 Derived ratios across constraint cases.

	$M=M$, $R_1=\frac{4}{3}r_s$	$M=M$	$R=R$	$\rho=\rho$	$I=I$	**General**	**Classical ($\rho=\rho$)**
$\sqrt{D_1/D_2}$	2.24716e+2	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	2.24716e+2	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	2.24716e+2	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	2.24716e+2	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(m_1 r_2)/(m_2 r_1)$	5.04972e+4	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	5.04972e+4	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(r_2/r_1)^3$	5.04972e+4	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
GTD$_{s,1}$/GTD$_{s,2}$	1-4.99996e-1	1-9.00776e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
GTD$_{d,1}$/GTD$_{d,2}$	1-4.99997e-1	1-9.00780e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
$\sqrt{1-dtd_1^2}/\sqrt{1-dtd_2^2}$	1-4.99997e-1	1-9.00780e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
I_1/I_2	3.92163e-10	1.83673e-11	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	1.98031e-5	4.28571e-6	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	1.31430e-2	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	1.31430e-2	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
m_1/m_2	1.00000e+0	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
r_1/r_2	1.98031e-5	4.28571e-6	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{r_1/r_2}$	4.45007e-3	2.07020e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0